Knowledge organiser · climate action and evidence

Today I can

I can distinguish mitigation from adaptation.

I can connect an action to a scientific mechanism.

I can choose a baseline and repeated measure before making a claim.

Mitigation means reduce a cause. Adaptation means cope with an effect. Both can be useful. They do different jobs.

An action needs a mechanism: say how each step could help. Where fossil fuels supply energy, lower demand can mean fewer emissions.

Safe shade can cut heat harm. That shade mechanism does not directly cut the cause of warming.

A baseline is what happens before. Use the same measure before and after. Repeat it. Keep conditions comparable.

The invented light-minute totals are 300 before and 90 after. They show less wasted-light time in that sample. They do not measure electricity, carbon or a global climate result.

Plan first. Keep one limit in view. A poster or a promise alone does not show success. Use approved paper scenarios. No live site action is needed.

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Word help · keep the science and a plain bridge

Today I can

I can distinguish mitigation from adaptation.

I can connect an action to a scientific mechanism.

I can choose a baseline and repeated measure before making a claim.

Mitigation: reduce a cause of climate change. Adaptation: reduce harm from an effect.

Mechanism: the link that explains how an action could help. Say the steps in order.

Baseline: a measure taken before a change. Repeated measure: check more than once with the same rule.

Comparable: fair to compare. Keep the same lamps and conditions when you compare light-on minutes.

Emissions: gases released into the air. Carbon dioxide is one greenhouse gas.

Confounding factor: another change that could affect the result. Provisional: a claim that may change with new evidence.

Light-on minutes: time a lamp is on. It is not a reading of energy or carbon. Keep that limit clear.

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Arrival · Supported

Today I can

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Use your book. Point, speak, sign, draw, write or direct an adult. An adult may read or scribe your exact words.

Use this recap. Extra greenhouse gases add to warming.

A finding needs useful evidence that we can trace.

Compare fairly. Take more than one measure.

Before saying an action worked, what should we measure?

1. Point to the action that cuts wasted energy.
2. Does shade cut heat exposure: yes or no?
3. Match before to baseline.
4. Finish: we measure before and ___.

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Arrival · Independent

Today I can

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Use your book. Point, speak, sign, draw, write or direct an adult. An adult may read or scribe your exact words.

Use this recap. Extra greenhouse gases add to warming.

A finding needs useful evidence that we can trace.

Compare fairly. Take more than one measure.

Before saying an action worked, what should we measure?

1. How could turning off an unused light cut emissions?
2. What is one difference between mitigation and adaptation?
3. Name one measure needed before the action starts.
4. Why is a poster alone not evidence of impact?

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Arrival · Aspire

Today I can

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I can choose a baseline and repeated measure before making a claim.

Use your book. Point, speak, sign, draw, write or direct an adult. An adult may read or scribe your exact words.

Use one link at a time. Name what changes. Keep the claim local and say what the data cannot show.

Use this recap. Extra greenhouse gases add to warming.

A finding needs useful evidence that we can trace.

Compare fairly. Take more than one measure.

Before saying an action worked, what should we measure?

1. Give one mitigation and one adaptation for different problems.
2. Link an action to energy demand, then to possible emissions.
3. Name one thing to keep comparable before and after.
4. Why can a local result not prove a global climate outcome?

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Starter model 1 · PROBLEM · name it

Today I can

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Name a cause and an effect. Extra greenhouse gases add to warming. Heat and rain impacts can differ from place to place. This is a simple picture, not a local forecast.

Look at one link at a time. Say what could cause the next step. Ask an adult to read the labels if needed.

PROBLEM

Extra greenhouse gases → warmer average

→ hotter summers, heavier rain here

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Starter model 2 · ACTION · explain the mechanism

Today I can

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Compare two kinds of action. Cutting needless electricity demand can cut emissions where fossil fuels supply energy. Safe shade can lower heat harm. That is coping with an effect.

These are useful but different jobs. Point to the cause-cutting action, then to the action that helps people cope. Use the labels to give a reason.

REDUCE the cause	COPE with the effect
switch off unused lights	shade sails on the yard
less energy → less fossil fuel	lowers heat exposure
→ less CO_2	does not lower CO_2

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Starter model 3 · MEASURE · compare and review

Today I can

I can distinguish mitigation from adaptation.

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I can choose a baseline and repeated measure before making a claim.

Plan the measure before the action. Use the same instrument at the same time of day. Repeat before and after under fair conditions. Staff supply any real readings.

A poster alone is not a measure. A repeated comparison can support a limited local claim. It does not prove measured energy, carbon or global climate success.

MEASURE

before: meter reading / temperature log

after: same meter, same time, repeated

did it change? by how much?

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Starter · choose a first idea

Today I can

I can distinguish mitigation from adaptation.

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I can choose a baseline and repeated measure before making a claim.

Use your book. Point, speak, sign, draw, write or direct an adult. An adult may read or scribe your exact words.

Which plan reduces a cause, and which plan helps people cope with an effect?

1. Switch off lights in empty rooms
2. Provide safe shade during high heat
3. Put up a poster and take no measurements

Commit one first idea before later frames. A first idea is held, not marked. Pointing, saying, signing or choosing all count.

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Model · link an action to how it helps

Today I can

I can distinguish mitigation from adaptation.

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I can choose a baseline and repeated measure before making a claim.

Problem: lights may be left on in empty rooms. This uses electricity that is not needed.

Action: only staff make an approved safe change. Where fossil fuels supply energy, lower demand can mean fewer emissions.

Contrast: safe shade cuts heat exposure. It is useful adaptation. It does not directly cut the cause of warming.

Before and after links a claim to what we can observe. A promise or poster is not a measure.

Compare the same lamps and conditions. Light-on minutes may suggest possible electricity savings. They do not measure energy, carbon or a global climate result.

Trace the link in order. Say what is used and what may go into the air. Keep the word can: the energy source matters.

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Action sorter · six source actions

Today I can

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I can choose a baseline and repeated measure before making a claim.

Use your book. Point, speak, sign, draw, write or direct an adult. An adult may read or scribe your exact words.

Choose an action, then choose reduces a CAUSE or copes with an EFFECT. Give a reason. Take turns as chooser and reason-questioner.

1. Turn off classroom lights and screens when empty
2. Plant trees for shade on the yard
3. Walk or cycle instead of a short car trip
4. Install a water butt to cope with dry spells
5. Insulate the building so less gas is burned
6. Move sports day earlier in the day to avoid heat

These are examples to discuss, not instructions to act. Use paper scenarios. Do not ask for anyone’s family travel or energy data.

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Action sorter · how each link could help

Today I can

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I can choose a baseline and repeated measure before making a claim.

1. Less electricity is needed. Where fossil fuels supply it, less fuel may be burned and fewer emissions released.

2. The named shade lowers heat exposure on hot days. That shade mechanism does not remove the extra carbon dioxide that caused the warming.

3. No petrol is burned for that trip, so it produces no carbon dioxide from burning petrol.

4. Rain is stored for dry weeks. It helps people cope; it does not change the climate.

5. Less heat is lost, so less gas is needed. Burning less gas releases less carbon dioxide.

6. This avoids the hottest hours. It is adaptation, rather than a direct cut in emissions.

Classify trees here by the named shade mechanism. This does not mean trees can never affect carbon dioxide.

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How would we measure it?

Today I can

I can distinguish mitigation from adaptation.

I can connect an action to a scientific mechanism.

I can choose a baseline and repeated measure before making a claim.

Use the same tool. Use it at the same time of day. Take more than one reading before and after. Keep the conditions fair.

The source examples are staff-supplied electricity meter readings, a heat log from a thermometer, or counts of car journeys. Staff supply real readings. No pupil uses live controls. No one shares family data.

One reading is too weak for this claim. A repeated fair check gives stronger evidence. Keep a limit as well.

Light-on minutes record time. They do not measure energy or carbon. They cannot prove a global climate effect. Keep the useful local result.

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We do · choose the plan that checks a result

Today I can

I can distinguish mitigation from adaptation.

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I can choose a baseline and repeated measure before making a claim.

Use your book. Point, speak, sign, draw, write or direct an adult. An adult may read or scribe your exact words.

Which plan can check whether the lighting action worked?

1. Record unused light-on minutes for five days before and five days after. Measure in the same way.

2. Put up a poster, then state that the problem is solved.

3. Ask one person whether the climate feels better.

Choose first. Use before, after and the same measure as clues. Explain or retry. An adult names back the science without upgrading your words.

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Model 2 · plan before claiming success

Today I can

I can distinguish mitigation from adaptation.

I can connect an action to a scientific mechanism.

I can choose a baseline and repeated measure before making a claim.

Baseline: record what happens before a change. Predict what should change and why.

Action: only staff make an approved safe change. Repeat the measure using the same rule.

Review: compare the evidence. Keep “not yet known” if it is too weak.

Before and after links a claim to what we can observe. A promise or poster is not a measure.

Compare the same lamps and conditions. Light-on minutes may suggest possible electricity savings. They do not measure energy, carbon or a global climate result.

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Before/After Evidence Lab · five cards

Today I can

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Use your book. Point, speak, sign, draw, write or direct an adult. An adult may read or scribe your exact words.

Sort each card as Mitigation, Adaptation or Evidence check. These invented data are for practice. They are not real school measurements.

The groups are MITIGATION / ADAPTATION / EVIDENCE CHECK.

1. Unused lights off: A staff-approved routine cuts electricity use that is not needed.

2. Safe shade in high heat: A shaded area cuts people’s exposure to heat.

3. Before: 300 minutes: Five invented daily records total 300 empty-room light-on minutes.

4. After: 90 minutes: Five invented daily records total 90 empty-room light-on minutes.

5. A poster proves success: A poster was put up. No behaviour or energy measure was collected.

Work as chooser and reason-questioner. Switch after two cards. Then each person explains one own choice.

Read one card first. Say what it checks. Then use the group clue and give a reason. You may try again.

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Evidence Lab · feedback and retry clues

Today I can

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I can choose a baseline and repeated measure before making a claim.

1. This is mitigation. It cuts a cause. Less needless energy demand can cut greenhouse-gas emissions where fuel is burned. Retry clue: Ask which job fits: cut a cause, cope with an effect, or check a result?

2. This is adaptation. It helps people cope with an effect. It does not directly remove the cause. Retry clue: Does this cut emissions or harm from heat? Use that clue to try again.

3. This is the baseline. It records what happened before the routine began. Retry clue: A number from before the action belongs in the evidence check.

4. Compared with the 300-minute baseline, wasted-light time fell in this invented sample. It is a local result. Retry clue: This measure checks what happened after the action.

5. This is an evidence check with a weak result. A poster does not show that impact changed. No useful measure was taken. Retry clue: Ask what was actually measured. A poster alone cannot show a change in impact.

Say only what this sample supports. Name one thing it cannot show. Keep the size of the claim fair.

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Invented data · compare five daily records

Today I can

I can distinguish mitigation from adaptation.

I can connect an action to a scientific mechanism.

I can choose a baseline and repeated measure before making a claim.

All figures are invented empty-room light-on minutes. They are for practice, not real school measurements.

Day 1: before 60; after 15. Day 2: before 45; after 20. Day 3: before 75; after 10.

Day 4: before 50; after 25. Day 5: before 70; after 20.

Before total: 60 + 45 + 75 + 50 + 70 = 300 minutes. After total: 15 + 20 + 10 + 25 + 20 = 90 minutes.

Compare the same lamps and conditions. This sample supports less wasted-light time here. It does not measure energy, carbon or global climate success.

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Science glitch · keep the useful local evidence

Today I can

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I can choose a baseline and repeated measure before making a claim.

Use your book. Point, speak, sign, draw, write or direct an adult. An adult may read or scribe your exact words.

Claim to test: Wasted-light time fell, so this one action fixed climate change.

1. Keep it: one local result proves the global outcome.

2. Repair it: the sample supports less wasted-light time here, not a global climate claim.

3. Erase the result because local evidence is never useful.

Choose first, then say what the data show and what they cannot show. A useful local result must not be erased.

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Approved school scenarios · plan only

Today I can

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Use your book. Point, speak, sign, draw, write or direct an adult. An adult may read or scribe your exact words.

Scenario A: staff-approved unused-lights routine. Plan on paper. Staff alone decide on any live change.

Scenario B: safe shade during high heat. Plan on paper. Staff approve the safe place and access needs.

Scenario C: water stored for dry spells. Use the source water-butt example as a paper plan. Pupils do not fit or operate equipment.

Supported chooses A or B. Other routes may choose any one approved scenario. No live change is required. Do not interrupt safety or access equipment. No family travel, food or energy data are needed.

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Independent · Supported

Today I can

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Use your book. Point, speak, sign, draw, write or direct an adult. An adult may read or scribe your exact words.

Pick one approved school scenario. Plan the action, how it helps and how to measure it. Do this before any claim of success.

Choose labelled cards for Action, How it helps and What we measure.

Choose between the unused-lights and safe-shade scenarios. Use a before/after picture strip. Say what to check before and after.

Use an Action / How it helps / What we measure chain. Include the baseline, repeated measure, a comparable condition and one limit.

Pupils do not touch sockets, plugs, boilers, thermostats or site controls. Staff must approve any action first.

Use your book, paper plan or cards. You may point or dictate exact words. No equipment or live site action is needed.

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Independent · Independent

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Use your book. Point, speak, sign, draw, write or direct an adult. An adult may read or scribe your exact words.

Pick one approved school scenario. Plan the action, how it helps and how to measure it. Do this before any claim of success.

Make the full plan. Include a baseline, prediction, repeated measure and one condition to keep the same.

Use: If we ___, then ___ because ___. We will measure ___. Keep a fair test and one limit in view.

Use an Action / How it helps / What we measure chain. Include the baseline, repeated measure, a comparable condition and one limit.

Pupils do not touch sockets, plugs, boilers, thermostats or site controls. Staff must approve any action first.

Use your book, paper plan or cards. You may point or dictate exact words. No equipment or live site action is needed.

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Independent · Aspire

Today I can

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I can choose a baseline and repeated measure before making a claim.

Use your book. Point, speak, sign, draw, write or direct an adult. An adult may read or scribe your exact words.

Pick one approved school scenario. Plan the action, how it helps and how to measure it. Do this before any claim of success.

Compare mitigation with adaptation. Name one confounding factor or evidence limit. A confounding factor is another change that could affect the result.

Explain why the result must stay local and provisional. That means it may change when new evidence is collected. Do not claim success before then.

Use an Action / How it helps / What we measure chain. Include the baseline, repeated measure, a comparable condition and one limit.

Pupils do not touch sockets, plugs, boilers, thermostats or site controls. Staff must approve any action first.

Use your book, paper plan or cards. You may point or dictate exact words. No equipment or live site action is needed.

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Review · next question, real audience, next move

Today I can

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Use your book. Point, speak, sign, draw, write or direct an adult. An adult may read or scribe your exact words.

Check your action, how it helps and the measure. Is it safe? Could the before and after be compared fairly? Keep one limit in view.

First question card for the next lesson. The safe question I choose for the next lesson is _____. We will measure _____.

Keep this pupil-owned card with the lesson evidence. The next lesson starts from the chosen question.

Show or say your own response to the adult. Hear the science they received. The adult must keep your meaning exact.

Then choose a named next move: secure / mixed / misconception / access / method or evidence. Say or show “Next I will ___.” Keep what changed after the adult listened.

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Exit · Supported

Today I can

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Use your book. Point, speak, sign, draw, write or direct an adult. An adult may read or scribe your exact words.

Mitigation or adaptation: switching off an unused light?

Mitigation cuts a cause. Adaptation helps people cope with an effect. Keep each link clear.

Use one fact or model clue to explain. A poster or a promise alone cannot show impact. Keep what is not yet known.

The adult hears or sees your own response, names back its science and agrees one next move. Point, speak, sign, draw, write or use exact scribing.

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Exit · Independent

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Use your book. Point, speak, sign, draw, write or direct an adult. An adult may read or scribe your exact words.

Explain one action-to-mechanism chain. Say how the action could help.

Mitigation cuts a cause. Adaptation helps people cope with an effect. Keep each link clear.

Use one fact or model clue to explain. A poster or a promise alone cannot show impact. Keep what is not yet known.

The adult hears or sees your own response, names back its science and agrees one next move. Point, speak, sign, draw, write or use exact scribing.

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Exit · Aspire

Today I can

I can distinguish mitigation from adaptation.

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I can choose a baseline and repeated measure before making a claim.

Use your book. Point, speak, sign, draw, write or direct an adult. An adult may read or scribe your exact words.

Name the measure and one limit. Why can success not be claimed before the evidence is collected?

Mitigation cuts a cause. Adaptation helps people cope with an effect. Keep each link clear.

Use one fact or model clue to explain. A poster or a promise alone cannot show impact. Keep what is not yet known.

The adult hears or sees your own response, names back its science and agrees one next move. Point, speak, sign, draw, write or use exact scribing.

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